Question 7 - Jenkinsfile pipeline stages

Question 7: Create a Jenkinsfile defining pipeline stages (build, test, deploy) and demonstrate a declarative or scripted pipeline execution.

Repository: https://github.com/ksamirdev/ensp461-devops-lab1

Jenkinsfile (declarative pipeline)

```
pipeline {
    agent any
    options { disableConcurrentBuilds() }
    triggers { pollSCM('H/2 * * * *') }

    stages {
        stage('Build') {
            steps {
                sh 'python3 -m venv .venv'
                sh '.venv/bin/pip install -q -r requirements.txt'
            }
        }
        stage('Test') {
            steps {
                sh '.venv/bin/python -m pytest -v --junitxml=test-results.xml'
            }
            post {
                always { junit 'test-results.xml' }
            }
        }
        stage('Package') {
            steps {
                sh 'mkdir -p dist && tar czf dist/text-stats-${BUILD_NUMBER}.tar.gz service requirements.txt'
                archiveArtifacts artifacts: 'dist/*.tar.gz', fingerprint: true
            }
        }
        stage('Deploy') {
            steps {
                sh 'sudo /usr/local/bin/text-stats-deploy "$WORKSPACE/dist/text-stats-${BUILD_NUMBER}.tar.gz"'
            }
        }
    }

    post {
        success { echo "Build ${BUILD_NUMBER} deployed" }
        failure { echo "Build ${BUILD_NUMBER} failed, deployed release unchanged" }
    }
}
```

Stages

Stage	Action	Fails when
Build	Creates a venv and installs requirements	A dependency cannot be installed
Test	Runs pytest and publishes JUnit results	Any test fails
Package	Creates and archives a tarball	Archive creation fails
Deploy	Installs the tarball, restarts the service, checks /healthz	Health check fails after 15 attempts

Execution

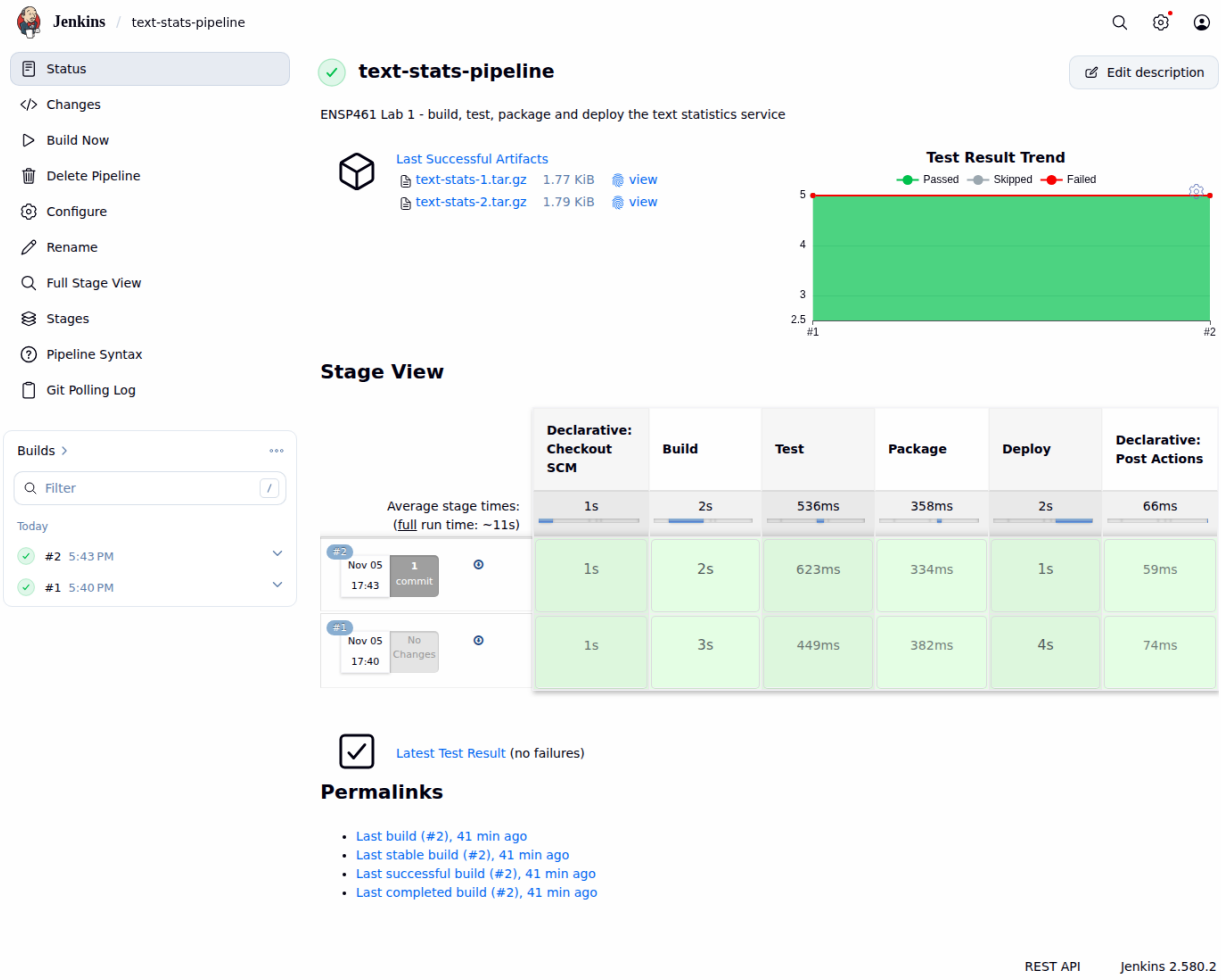


Fig: Stage view for the pipeline

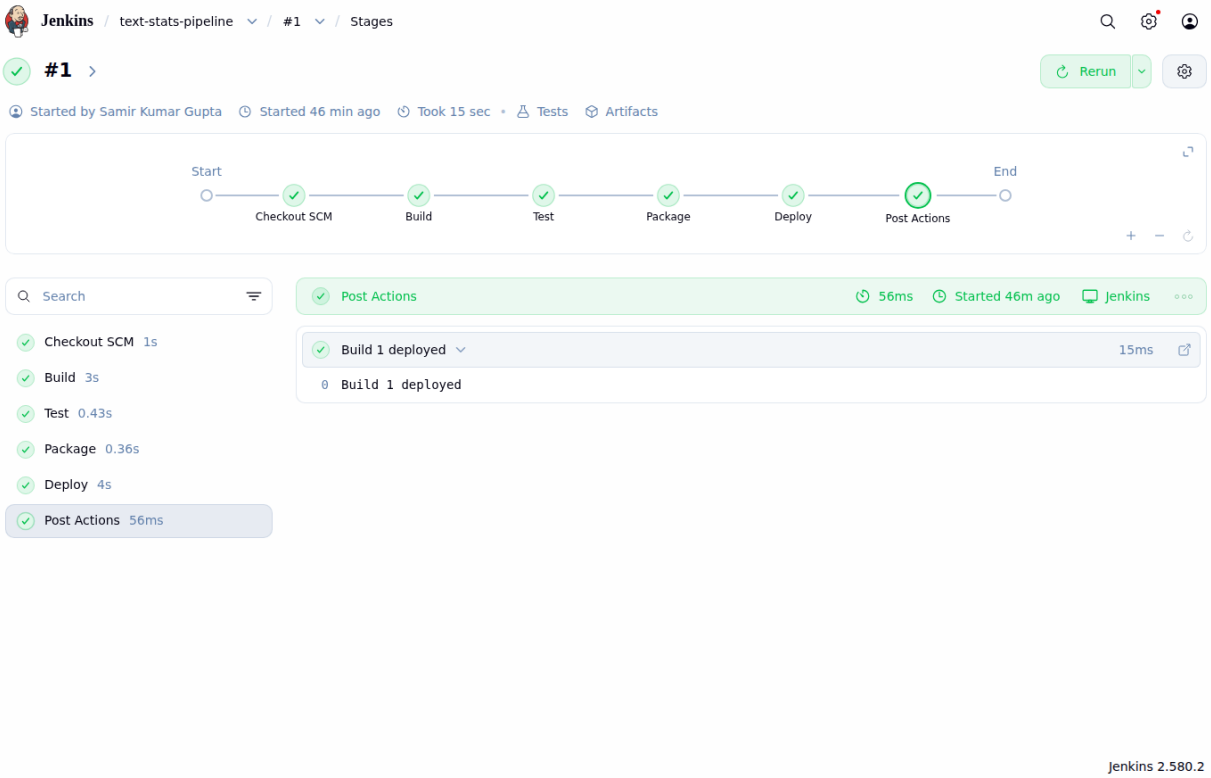


Fig: Pipeline graph of build 1

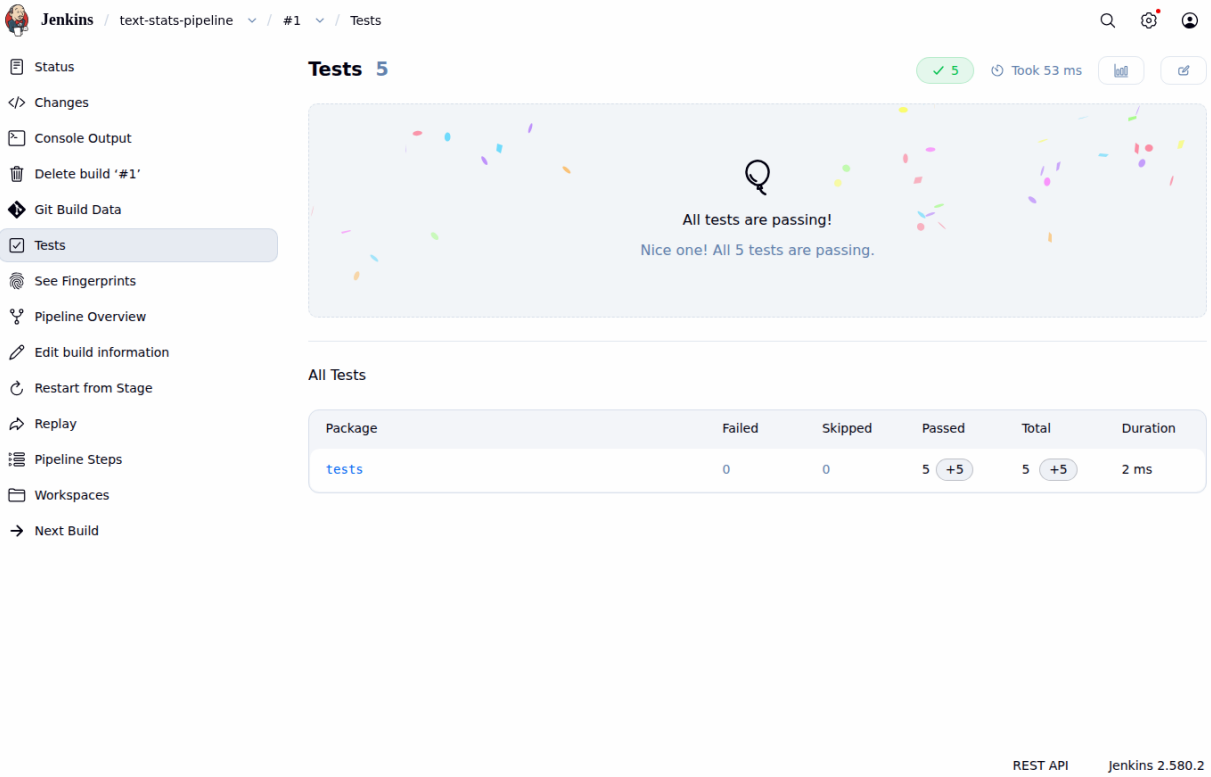


Fig: Test results of build 1

Observations

- Declarative stages make failures easy to locate because each stage is reported separately.
- The junit step publishes test results even when tests fail because it runs in post { always }.
- In build 1 the first health probe failed while the service was starting, and the retry loop in the deploy script passed on a later attempt.